

Task 1: My World as an MDP

Recall-E · Week 1

Scenario: The Student's Daily Choice

Every morning a student decides whether to Study, Sleep, or Hang Out — a choice that depends on their current energy and how behind they are academically. This maps naturally onto an MDP.

States & Actions

State	Meaning	Action	Meaning
s0	Tired & Behind	a0	Study
s1	Balanced	a1	Sleep
s2	Energised	a2	Hang Out

Transition Probabilities $P(s' | s, a)$

(s, a)	$P(s'=s0)$	$P(s'=s1)$	$P(s'=s2)$
s0, Study	0.5	0.4	0.1
s0, Sleep	0.1	0.6	0.3
s0, Hang Out	0.6	0.3	0.1
s1, Study	0.2	0.5	0.3
s1, Sleep	0.0	0.4	0.6
s1, Hang Out	0.2	0.5	0.3
s2, Study	0.1	0.3	0.6
s2, Sleep	0.0	0.2	0.8
s2, Hang Out	0.0	0.3	0.7

Reward Function $R(s, a)$

(s, a)	R	Why
s0, Study	+1	Progress despite fatigue
s0, Sleep	+3	Needed rest
s0, Hang Out	-2	Wasted time while behind
s1, Study	+4	Productive and balanced
s1, Sleep	+1	Not really needed
s1, Hang Out	+2	Healthy balance
s2, Study	+5	Peak productivity
s2, Sleep	-1	Already well-rested

s2, Hang Out	+3	Guilt-free social time
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Value Iteration ($\gamma = 0.9$, two steps)

Start with $V_0(s) = 0$. Update rule: $V^{k+1}(s) = \max_a [R(s,a) + \gamma \cdot \sum P(s'|s,a) \cdot V^k(s')]$

Step 1 $\rightarrow V_1$ ($V_0 = 0$, so just pick the best immediate reward):

$$V_1(s_0) = \max(1, 3, -2) = 3 \text{ [Sleep]}$$

$$V_1(s_1) = \max(4, 1, 2) = 4 \text{ [Study]}$$

$$V_1(s_2) = \max(5, -1, 3) = 5 \text{ [Study]}$$

Step 2 $\rightarrow V_2$ (now the future look-ahead kicks in):

$$Q(s_0, \text{Study}) = 1 + 0.9 \times (0.5 \times 3 + 0.4 \times 4 + 0.1 \times 5) = 1 + 3.24 = 4.24$$

$$Q(s_0, \text{Sleep}) = 3 + 0.9 \times (0.1 \times 3 + 0.6 \times 4 + 0.3 \times 5) = 3 + 3.78 = 6.78 \checkmark$$

$$Q(s_0, \text{HangOut}) = -2 + 0.9 \times (0.6 \times 3 + 0.3 \times 4 + 0.1 \times 5) = -2 + 3.15 = 1.15$$

$$\rightarrow V_2(s_0) = 6.78 \text{ [Sleep]}$$

$$Q(s_1, \text{Study}) = 4 + 0.9 \times (0.2 \times 3 + 0.5 \times 4 + 0.3 \times 5) = 4 + 3.69 = 7.69 \checkmark$$

$$Q(s_1, \text{Sleep}) = 1 + 0.9 \times (0.0 \times 3 + 0.4 \times 4 + 0.6 \times 5) = 1 + 4.14 = 5.14$$

$$Q(s_1, \text{HangOut}) = 2 + 0.9 \times (0.2 \times 3 + 0.5 \times 4 + 0.3 \times 5) = 2 + 3.69 = 5.69$$

$$\rightarrow V_2(s_1) = 7.69 \text{ [Study]}$$

$$Q(s_2, \text{Study}) = 5 + 0.9 \times (0.1 \times 3 + 0.3 \times 4 + 0.6 \times 5) = 5 + 4.05 = 9.05 \checkmark$$

$$Q(s_2, \text{Sleep}) = -1 + 0.9 \times (0.0 \times 3 + 0.2 \times 4 + 0.8 \times 5) = -1 + 4.32 = 3.32$$

$$Q(s_2, \text{HangOut}) = 3 + 0.9 \times (0.0 \times 3 + 0.3 \times 4 + 0.7 \times 5) = 3 + 4.23 = 7.23$$

$$\rightarrow V_2(s_2) = 9.05 \text{ [Study]}$$

State	V0	V1	V2	Best Action
s0 (Tired)	0	3	6.78	Sleep
s1 (Balanced)	0	4	7.69	Study
s2 (Energised)	0	5	9.05	Study

Takeaway: A tired student should sleep first — rest today pays off tomorrow. Once balanced or energised, studying is always optimal. Values will keep growing with more iterations until convergence.